TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	77.5 ± 3.26
Correlation Stability	87.9 ± 2.12
Distribution Consistency	86.1 ± 3.01
Entropy Preservation	87.6 ± 1.36
Mean Z Score Drift	0.2 ± 0.06
Categorical Fidelity	83.4 ± 0.49
Privacy Risk	49.7 ± 1.55
Data Density	99.9 ± 0.21
Coverage Score	18.6 ± 5.14
Feature Accuracy	77.4 ± 4.34
Novelty Score	0.5 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	39.491	-5.255	11.7%	Warn
TB	3.299	1.785	-1.513	45.9%	Drift
DB	1.486	0.749	-0.737	49.6%	Drift
Alkphos	290.576	248.841	-41.735	14.4%	Warn
Sgpt	80.714	49.142	-31.572	39.1%	Drift
Sgot	109.911	54.663	-55.248	50.3%	Drift
TP	6.483	6.733	+0.250	3.9%	OK
ALB	3.142	3.401	+0.259	8.2%	OK
A/G Ratio	0.947	1.040	+0.092	9.8%	OK
Selector	1.286	1.330	+0.044	3.4%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	77.5 ± 3.26
Correlation Stability (*2)	87.9 ± 2.12
Distribution Consistency (*3)	86.1 ± 3.01
Entropy Preservation (*4)	87.6 ± 1.36
Mean Z Score Drift (*5)	0.2 ± 0.06
Categorical Fidelity (*6)	83.4 ± 0.49
Privacy Risk (*7)	49.7 ± 1.55
Data Density (*8)	99.9 ± 0.21
Coverage Score (*9)	18.6 ± 5.14

Feature Accuracy (*10)	77.4 ± 4.34
Novelty Score (*11)	0.5 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.558974358974359	0.5512820512820513	0.5692307692307692	0.010256410256410243	
F1	0.5587432119380917	0.6361728876239088	0.5775283532857696	0.018785141347677912	*

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.5735042735042735	0.6606837606837608	-0.025641025641025654	
F1	0.7971299319768493	0.6867788218944428	0.7799634903478443	-0.017166441629005026	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.5333333333333334	0.7017094017094017	-0.004273504273504274	
F1	0.8128002697439015	0.6351051924260909	0.8106725177376752	-0.002127752006226284	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5717948717948718	0.6914529914529914	-0.011111111111111117	
F1	0.7997903571549948	0.6693012961991565	0.789170410927037	-0.010619946227957732	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7	0.5316239316239317	0.6957264957264957	-0.004273504273504274	
F1	0.8064863225628688	0.628234767499898	0.8026491198601434	-0.003837202702725406	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6846153846153846	0.5538461538461539	0.7017094017094017	0.01709401709401709	
F1	0.7897853358907824	0.6599683117337967	0.7998278226113366	0.010042486720553656	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7017094017094017	0.5820512820512821	0.7111111111111111	0.009401709401709379	
F1	0.8077538159701655	0.6916111462520309	0.8132236795127944	0.005469863542628284	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6213675213675214	0.7068376068376068	0	
F1	0.8279091624529537	0.7538305682217745	0.8279091624529537	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5769230769230769	0.6923076923076923	0.01794871794871795	
F1	0.7821343746683586	0.6799823070082086	0.7963222920478272	0.014187917379468654	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6538461538461543	0.5153846153846153	0.6606837606837607	0.006837606837606825	
F1	0.7592392809534111	0.6096454427549799	0.7600174699975959	0.000778189044184840 5	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	16	Male	2.6	1.2	236	131	90	5.4	2.6	0.9
2	32	Male	15.9	7.0	280	1350	1600	5.6	2.8	1.0
3	30	Female	0.8	0.2	198	30	58	5.2	2.8	1.1
4	66	Female	2.9	1.3	168	21	38	5.5	1.8	0.4
5	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
6	75	Male	0.9	0.2	282	25	23	4.4	2.2	1.0
7	25	Female	1.2	0.5	210	45	50	6.5	3.2	0.9
8	34	Male	0.8	0.2	190	30	40	6.0	2.8	1.1
9	29	Female	2.5	1.0	250	60	70	7.0	3.5	1.0
10	41	Male	3.0	1.5	300	80	90	7.5	3.8	1.2
11	55	Female	0.9	0.3	175	25	35	5.8	2.6	1.1
12	38	Male	1.4	0.6	220	50	60	6.7	3.1	0.8
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	20	Male	1.1	0.5	128	20	30	3.9	1.9	0.95
2	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
3	33	Male	1.2	0.3	498	28	25	7.0	3.0	0.7
4	64	Male	1.1	0.5	145	20	24	5.5	3.2	1.39
5	69	Female	0.8	0.2	146	42	70	8.4	4.9	1.4
6	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
7	22	Female	0.9	0.3	150	25	30	6.0	3.2	1.0
8	34	Male	1.5	0.4	200	40	50	7.0	3.5	0.9
9	45	Female	2.1	0.5	250	30	40	6.5	3.0	1.1
10	29	Male	1.2	0.3	180	20	25	5.8	2.8	0.8
11	55	Female	0.8	0.2	160	35	45	7.2	3.4	1.2
12	37	Male	1.0	0.4	190	28	38	6.3	3.1	0.9
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	20	Female	16.7	8.4	200	91	101	6.9	3.5	1.02
2	66	Female	4.2	2.1	159	15	30	7.1	2.2	0.4
3	26	Male	7.1	3.3	258	80	113	6.2	2.9	0.8
4	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
5	28	Female	1.0	0.3	90	18	108	6.8	3.1	0.8
6	32	Male	15.6	9.5	134	54	125	5.6	4.0	2.5
7	29	Female	1.5	0.7	210	45	50	6.5	3.0	0.6
8	34	Male	0.9	0.4	180	30	40	5.8	2.5	0.7
9	41	Female	2.3	1.1	300	50	60	7.0	3.8	1.0
10	50	Male	3.5	1.8	250	70	80	6.2	3.1	0.9
11	22	Female	0.6	0.3	190	20	30	5.5	2.0	0.5
12	38	Male	1.2	0.5	220	40	50	6.8	3.4	1.1
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	23	Male	1.1	0.5	191	37	41	7.7	4.3	1.2
2	56	Male	1.1	0.5	180	30	42	6.9	3.8	1.2
3	50	Male	0.7	0.2	206	18	17	8.4	4.2	1.0
4	26	Female	0.7	0.2	144	36	33	8.2	4.3	1.1
5	66	Female	0.7	0.2	162	24	20	6.4	3.2	1.0
6	46	Female	1.4	0.4	298	509	623	3.6	1.0	0.3
7	34	Female	0.9	0.3	150	30	40	6.5	3.2	1.0
8	29	Male	1.2	0.5	200	45	50	7.0	4.0	1.1
9	52	Female	0.8	0.2	180	25	30	6.0	3.5	0.9
10	41	Male	1.5	0.7	220	60	70	7.5	4.2	1.2
11	38	Female	0.6	0.1	160	20	25	6.8	3.0	0.8
12	45	Male	2.0	0.9	250	80	90	7.8	4.5	1.3
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	39	Male	3.8	1.5	298	102	630	7.1	3.3	0.8
2	20	Female	0.6	0.2	202	12	13	6.1	3.0	0.9
3	42	Female	2.3	1.1	292	29	39	4.1	1.8	0.7
4	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
5	54	Female	22.6	11.4	558	30	37	7.8	3.4	0.8
6	22	Male	2.4	1.0	340	25	21	8.3	4.5	1.1
7	34	Female	1.2	0.5	210	22	30	6.5	3.2	1.0
8	29	Male	0.9	0.3	180	25	35	7.0	3.5	1.1
9	50	Female	2.5	1.0	250	30	40	7.5	3.8	1.2
10	45	Male	1.8	0.7	220	28	32	6.8	3.4	1.1
11	55	Female	0.7	0.2	190	20	25	7.2	3.0	0.9
12	62	Male	3.0	1.2	300	35	50	8.0	4.0	1.3
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	66	Male	0.7	0.2	239	27	26	6.3	3.7	1.4
2	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
3	22	Male	0.6	0.2	202	78	41	8.0	3.9	0.9
4	50	Male	0.6	0.2	137	15	16	4.8	2.6	1.1
5	31	Female	0.8	0.2	215	15	21	7.6	4.0	1.1
6	25	Female	0.7	0.1	140	32	25	7.6	4.3	1.3
7	34	Female	1.2	0.5	150	25	30	6.5	3.0	1.1
8	29	Male	0.9	0.3	200	20	22	7.0	3.5	1.2
9	55	Female	0.8	0.2	180	35	40	7.2	3.8	1.3
10	42	Male	2.1	0.9	250	60	70	6.8	3.2	0.9
11	38	Female	1.5	0.4	220	28	35	7.5	4.0	1.0
12	60	Male	3.0	1.2	300	80	90	6.7	2.9	0.8
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	21	Male	18.5	9.5	380	390	500	8.2	4.1	1.0
2	47	Male	0.9	0.2	265	40	28	8.0	4.0	1.0
3	55	Male	1.1	0.3	215	21	15	6.2	2.9	0.8
4	60	Male	8.6	4.0	298	412	850	7.4	3.0	0.6
5	42	Male	30.5	14.2	285	65	130	5.2	2.1	0.6
6	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
7	29	Female	1.2	0.4	150	35	40	6.5	3.0	0.8
8	34	Male	2.1	0.9	200	45	50	7.0	3.5	0.7
9	41	Female	0.8	0.3	180	30	25	6.0	2.8	0.9
10	55	Male	5.5	2.0	300	60	70	7.8	3.6	0.6
11	62	Female	1.5	0.5	220	40	30	6.2	3.1	1.0
12	38	Male	0.9	0.2	190	20	35	5.5	2.7	0.8
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
2	20	Female	16.7	8.4	200	91	101	6.9	3.5	1.02
3	39	Male	1.9	0.9	180	42	62	7.4	4.3	1.38
4	50	Male	0.8	0.2	152	29	30	7.4	4.1	1.3
5	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
6	38	Male	0.7	0.2	110	22	18	6.4	2.5	0.64
7	34	Female	0.9	0.3	150	45	30	6.5	3.2	1.1
8	29	Male	1.2	0.4	200	50	40	7.0	3.5	1.2
9	42	Female	0.8	0.2	180	38	35	6.8	3.4	1.0
10	55	Male	2.5	1.0	250	60	55	7.5	4.0	1.3
11	48	Female	1.0	0.5	220	55	45	6.9	3.8	1.1
12	37	Male	0.7	0.2	160	40	30	6.4	2.9	0.9
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	18	Male	1.4	0.6	215	440	850	5.0	1.9	0.6
2	39	Male	3.8	1.5	298	102	630	7.1	3.3	0.8
3	27	Male	1.2	0.4	179	63	39	6.1	3.3	1.1
4	60	Male	0.8	0.2	286	21	27	7.1	4.0	1.2
5	31	Male	0.8	0.2	198	43	31	7.3	4.0	1.2
6	13	Female	0.7	0.1	182	24	19	8.9	4.9	1.2
7	25	Female	1.0	0.3	150	35	45	6.5	3.5	1.2
8	45	Male	2.5	1.0	300	80	120	7.0	4.0	1.5
9	34	Female	0.9	0.2	200	50	60	6.8	3.2	1.1
10	29	Male	1.2	0.5	180	40	55	6.3	3.0	1.0
11	52	Female	3.0	1.2	250	70	90	7.5	4.5	1.3
12	38	Male	0.8	0.3	220	45	70	6.7	3.8	1.1
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	24	Female	0.9	0.2	195	40	35	7.4	4.1	1.2
2	26	Female	0.6	0.2	142	12	32	5.7	2.4	0.75
3	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62
4	43	Female	0.9	0.3	140	12	29	7.4	3.5	1.8
5	60	Male	11.0	4.9	750	140	350	5.5	2.1	0.6
6	43	Male	0.8	0.2	192	29	20	6.0	2.9	0.9
7	29	Female	1.2	0.5	210	30	25	6.5	3.5	1.1
8	34	Male	0.9	0.3	180	45	50	7.0	3.0	1.2
9	40	Female	2.1	1.0	250	60	70	5.9	2.8	0.8
10	55	Male	3.5	1.8	300	80	90	6.2	3.1	1.0
11	22	Female	0.7	0.2	190	20	15	5.5	2.5	1.3
12	38	Male	1.8	0.7	220	35	40	6.8	3.4	0.9
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3